CREWREADY

BEYOND US

PRODUCT, GAMEPLAY & BUILD PLAN

NASA Space Apps Challenge 2026

Build a Junior Astronaut Mission Trainer

Explore space. Solve the impossible. Shape our future.

Start Here: How to Use This Document

This is the product source of truth for the team: the problem, solution, core loop, retention plan, feature tiers, PRD, UI/UX, technical architecture, build tasks, video script and judging strategy. Read and execute it in the order below. The wider vision is ambitious; the MVP is deliberately narrow. If two statements ever conflict, follow the exact challenge text first and the most specific implementation detail second.

NON-NEGOTIABLE SCOPE RULE

The competition build must be a working, testable mission that directly answers one official 2026 challenge. The broader product vision is not a promise to build every feature during the hackathon.

BUILD THIS FIRST

A polished one-tap Quick Run: a short Mars spacecraft mission in which real engineering obstacles (radiation, debris, air, power and communication delay) force trade-offs between the same four outpost systems, update the craft visibly, explain the science with sources and offer an instant retry. Then connect it to Mission 01, the Mars outpost. Add the campaign and multiplayer only after that path is reliable.

Priority rule: one complete, science-grounded outpost mission first. Add breadth only after the core mission works reliably.

Part	What it contains	Use it to
1. The Mission	The 2026 theme, the exact challenge, the problem, our solution and why Mars comes first.	Understand why we are building this.
2. Why We Can Win	Lessons from previous winners and the judging score sheet with the evidence we must show.	Aim every decision at the score sheet.
3. Product Concept and Gameplay	Product definition, one-tap home, Quick Run, Mission 01, campaign, survival model, twists, upgrades, training and crew play.	Know exactly what the player does.
4. MVP, Tiers and Priorities	Feature tiers, build priorities and what is deliberately left out.	Decide what is built now and what waits.
5. Product Requirements and Design	PRD, user stories, metrics, UI/UX system, screens and onboarding.	Design and accept each feature.
6. Technical Requirements	Architecture, game-state model, simulation rules, NASA data plan and child safety.	Build the engine and keep the science honest.
7. Build Plan	Step-by-step build order, workstreams, roles and test plan.	Assign owners and finish on time.
8. Submission	Competition calendar, demo, video scripts, pitch and final output checklist.	Present and submit with confidence.
9. Risks and Safeguards	Risks and the design fix for each.	Prevent avoidable failures.
10. Future Roadmap	Space Reels, creator ecosystem and long-term platform ideas.	Show vision without overpromising.

Part	What it contains	Use it to
11. Research and Sources	Official and research references.	Cite correctly.
12. Conclusion	Why CrewReady can win and the final product principle.	Stay focused.

1. The Mission: Theme, Challenge and Solution

1.1 The 2026 theme: The Next Frontier

The official 2026 theme is “The Next Frontier.” NASA frames it as discovery from our own backyard to the edges of the universe, turning data into tools that benefit humanity and help propel exploration to the Moon, Mars, and beyond. NASA also identifies three broad focus areas for this year: returning humans to the Moon, preparing for missions deeper into the solar system, and expanding our understanding of the universe.

Source: https://www.spaceappschallenge.org/2026/ and https://science.nasa.gov/universe/nasa-welcomes-all-ages-experience-levels-for-space-apps-challenge/

What this means for CrewReady: the theme is an invitation, not a substitute for a specific challenge. The team must name and solve one official 2026 challenge. CrewReady should only be submitted if its core mission directly answers that challenge, not simply because it is a space-themed game. Keep the selected challenge title and its exact problem statement visible on the project page, in the demo, and in the README.

1.2 The actual challenge

NASA Space Apps is a global hackathon built around challenge statements authored by NASA subject-matter experts. Teams create solutions using science, technology, storytelling and, where relevant, NASA or partner open data. It is not simply a competition to make the most entertaining space app.

> **SELECTED CHALLENGE**
> Build a Junior Astronaut Mission Trainer (NASA Space Apps 2026). CrewReady is a team concept, not an official NASA product or challenge statement. It answers this exact challenge. Use the official title exactly, quote its core ask in every submission item, and never imply NASA sponsorship or endorsement.

The official challenge asks for an interactive game or app in which students operate a lunar or Martian outpost and balance life support, radiation shielding, power, and food production. The goal is not simply to make space look exciting. Players must experience how engineering choices can improve one system while placing another at risk, and how those linked choices can determine mission success or failure.

CrewReady’s defining promise: “Every choice spends something. Every system affects another. Every mission teaches why.” The core experience is a playable outpost simulation, not a collection of unrelated mini-games. Story, twists, training, multiplayer, and reels must all support that core promise.

Why this solution addresses the challenge better

Challenge gap	CrewReady response	Proof in the playable demo
Trade-offs are often oversimplified	Every action changes a small, visible systems model. More shielding uses mass and construction time; extra lighting draws power; growing food consumes power, water, nutrients and crew time.	Show the before/after resource panel and explain at least two linked consequences.
Science can feel too complex for young learners	Use a readable control room, simple language, visual cause-and-effect, short explanations and optional “Mission Science” detail cards.	A first-time player can finish a short mission without a tutorial video.

Challenge gap	CrewReady response	Proof in the playable demo
Failure feels arbitrary in many games	Give players telemetry, warnings and partial clues. Failure follows understandable rules; debrief explains the causal chain.	Replay a failed run and show which decision caused the cascade.
Space survival is often only cosmetic	Model oxygen/air revitalisation, water recovery, power, food, radiation exposure, crew health and time as linked systems.	A system panel visibly updates after an action or event.
Education can stop after one play	Escalating mission contracts, new hazards, alternate solutions, cooperative roles and an optional creator feed extend learning.	A new mission changes the problem, not merely the background.

Challenge alignment checklist

- Select one exact official 2026 challenge and accurately quote its core ask in the submission. Selected: Build a Junior Astronaut Mission Trainer.
- Map each required outcome to a visible feature or evidence item.
- Use NASA or partner science/data where relevant to the selected challenge.
- Explain what the player learns or what problem the tool helps solve.
- The challenge allows a lunar or Martian outpost. Mars is the first setting; a Moon mission pack is planned later rather than building two shallow environments.
- Confirm current rules, allowed preparation, team size, media requirements and submission fields with the official site and Local Lead.

1.3 The problem we are solving

The user problem

Space exploration depends on connected systems: power, life support, food, communications, equipment and crew safety. Young learners may encounter these as separate facts or videos, but have fewer opportunities to make decisions, experience trade-offs and see how one system affects another.

The product problem

- A campaign that requires ten real-world days creates friction and drop-off.
- A passive library or endless video feed can hold attention without offering a distinctive learning experience.
- A complicated simulation can overwhelm players before they reach the fun.
- Multiplayer can fail if room setup, connections or role rules are confusing.
- Space-themed art alone does not demonstrate meaningful use of science or data.

> **PROBLEM STATEMENT**
>
> How might we help young people understand real space-exploration trade-offs through a game they can start in one tap, finish in minutes, replay endlessly, and optionally experience with friends?

1.4 Our solution

CrewReady: Beyond Us is a child-friendly space mission game. Players can launch an instant solo survival run, a short spacecraft mission, or work as a crew to operate a Mars outpost. They build a base with limited supplies, manage power and life-support systems, react to unexpected events, and learn from the outcome. Fast replayable runs sit alongside a campaign, upgrades and optional multiplayer. Science is sourced, simplified transparently and explained after key decisions.

CrewReady: Beyond Us is a child-friendly, science-grounded mission simulator that helps the next generation practise the thinking, teamwork, and physical-awareness skills needed to understand human exploration beyond Earth. Players take responsibility for a small lunar or deep-space outpost, make decisions under constrained resources, and see how those decisions affect crew safety and mission success. Real NASA open data and clearly sourced science turn each mission into a playable explanation, not just a fictional space adventure.

The product has three connected layers:

- **MISSION PLAY:** short cooperative outpost scenarios where players manage power, water, oxygen, communications, shelter, and crew priorities. A scenario must show why a choice matters and what evidence supports the outcome.
- **CREW TRAINING:** optional brain and body activities inspired by real astronaut-relevant capabilities, translated into safe, age-appropriate mini-games. These are educational simulations, not medical, fitness, or astronaut-selection tests.
- **DISCOVERY FEED:** a future horizontal swipe feed of short science explainers, NASA-data visualisations, creator experiments, and mission clips. The feed supports discovery and retention, but does not replace the core data-driven mission experience.

The strongest framing is not “a game that teaches kids about space.” It is “a data-informed mission learning tool that lets young people practise decisions about the systems that make exploration possible.” That distinction strengthens relevance, validity, impact, and future prospects.

Why this is a strong fit for the theme and future generations

Need / opportunity	CrewReady response	Why it matters
The Next Frontier	A mission moves from familiar Earth problems to lunar and deeper-space living, with the selected official challenge at its core.	Makes exploration tangible while staying connected to NASA’s stated Moon, solar-system, and universe focus areas.
Data must become useful tools	NASA datasets or NASA science resources power at least one visible mechanic, visualisation, or decision.	Judges can trace a line from source data to a player action and its consequence.
Alpha Generation and future learners	Fast starts, tactile choices, clear feedback, visual storytelling, cooperative roles, and optional short-form discovery.	Fits current interactive habits without assuming every child has the same attention span, device, bandwidth, or ability.
Learning by doing	Players predict, decide, observe outcomes, and retry with improved reasoning.	Makes science actionable instead of presenting a static information page.
Global access	Mobile-first layout, readable text, captions, low-bandwidth mode, and solo fallback.	Supports different devices, locations, language needs, and levels of connectivity.

Need / opportunity	CrewReady response	Why it matters
Long-term potential	Expandable mission packs, classroom mode, creator feed, and later creator rewards.	Creates a credible pathway from hackathon prototype to a wider educational platform, without overpromising the first build.

Why Mars for the first mission?

- Dust can reduce sunlight reaching solar panels, creating an intuitive power-management problem.
- Power, batteries, food production, life support and shelter create natural trade-offs that can be shown visually.
- One setting keeps the prototype coherent. A Moon mission can be added later rather than building two shallow environments.
- Verify all scenario facts against NASA sources and separate scientific facts from fictional gameplay values.
- Mars is chosen because dust storms are a documented Mars hazard for solar power. The Moon has different hazards (a lunar night of about two weeks and abrasive lunar dust), so a Moon mission pack comes later.

Designing for young learners: evidence, not stereotypes

“Gen Alpha” is a useful shorthand, not a single behaviour profile. The research below is mostly from the United States, so it should inform interface hypotheses rather than be presented as a direct measurement of Bangladeshi children. Test the prototype with local learners, including different ages, devices, internet access and levels of English/science confidence.

- Common Sense Media’s 2025 census of children aged 0–8 reported that gaming time had risen 65% over four years, short-form video was growing, and around one in three young children used AI for learning. This supports testing interactive learning and short visual explanations, not assuming every child wants the same format. Source: https://www.commonsensemedia.org/research/the-2025-common-sense-census-media-use-by-kids-zero-to-eight
- Pew Research Center’s 2024 survey reported that 90% of U.S. teens aged 13–17 used YouTube, while TikTok and Instagram were each used by roughly six in ten. This supports familiarity with visual discovery and video patterns, but does not prove that a reels feed improves learning. Source: https://www.pewresearch.org/wp-content/uploads/sites/20/2024/12/PI_2024.12.12_Teens-Social-Media-Tech_REPORT.pdf
- Pew’s 2024 study found that 85% of U.S. teens said they played video games and about four in ten played daily. This supports a game-based approach, while also reinforcing the need for accessible controls, healthy session lengths and a clear learning purpose. Source: https://www.pewresearch.org/internet/2024/05/09/teens-and-video-games-today/

How CrewReady acts on the evidence: start with a fast playable mission; teach through consequences instead of long lectures; use short optional explanations; give players meaningful choice; make progress visible; enable collaboration; support low-end phones and weak connections; and offer text/captions, audio-off play, reduced motion and a non-camera route. The project should be enjoyable without needing endless scrolling.

2. Why We Can Win: Lessons from Previous Winners and the Judging Score Sheet

2.1 What previous winners show

Winner projects do not reveal a guaranteed formula. Their public descriptions do show that strong submissions can combine a specific problem, meaningful science or data use, a tangible interactive experience and memorable communication. These are documented approaches and awards, not proof that any one feature caused a win.

Winner and award	Documented direction	Lesson	How CrewReady applies it
PureFlow, 2025 · Best Mission Concept	Interactive habitat design and simulation tested against space-weather threats.	Make base design and threat response part of the actual game loop. A credible habitat concept makes interconnected systems and threat testing tangible.	Let players change a resource or system choice and immediately see a clear consequence. Show how habitat decisions affect survival. Label which rules are sourced and which are simplified game assumptions.
HerCode Space, 2025 · Best Use of Storytelling	NASA space-weather concepts taught to children through story and illustration.	Use character, suspense and narrative to make science understandable and memorable.	Use a small cast, one understandable threat and a short mission narrative. Use character reports, mission logs and visual debriefs while keeping the science accurate.
Zumorroda-X, 2025 · Art & Technology	Mini-games about agriculture and responses to climate-related challenges.	Teach through doing, not only reading. Mini-games can make a data-driven topic approachable.	Use one well-chosen mini-game tied to an authentic NASA resource, not a collection of unrelated arcade mechanics. Use short training challenges only when they teach a mechanic used later in the mission.
Resonant Exoplanets, 2025 · Best Use of Data	AI applied to telescope data.	Technology should serve the data problem. A clear data pipeline and a specific analytical purpose make a strong technical story.	Only add AI if it materially helps interpret, explain or personalise a feature; a transparent rule-based system is better than decorative AI. Show which NASA source informs each model or display.
Twisters, 2025 · Best Use of Technology	Earth-observation data used to support practical weather decisions.	Turn data into a usable decision. Technology is strongest as a decision-support tool.	Make NASA data change the player's options, risk assessment or explanation; a citation list alone is not enough. Use telemetry and scenario warnings to support a better, explainable decision.
Eco-Metropolis (team Innovisionaries), 2024 · Most Inspirational	City-building game with resource decisions linked to environmental data.	Use progressive missions and meaningful resource trade-offs.	Escalate through missions that each add one new resource trade-off (see the campaign in Part 3).
Team Voyagers, 2023 · Best Storytelling	Interactive web game about the water cycle and climate impacts on freshwater.	A clear story helps users care about a scientific system.	Wrap the outpost systems in one clear mission story with Nova as the guide.

Bangladeshi teams have also reached global winner level: Team Olik (2018, Best Use of Data), Team Mohakash (2021, Best Mission Concept), Team Diamonds (2022, Most Inspirational) and Team Voyagers (2023, Best Storytelling).

The differentiator is disciplined integration. CrewReady should not attempt to beat mature research dashboards at scientific depth or compete with commercial games on content volume. The strongest strategy is to prove one memorable mission with a credible systems model, a clear learning outcome, purposeful NASA sourcing and an excellent presentation. A polished playable core is more persuasive than a broad but unfinished feature list.

Sources: https://www.nasa.gov/learning-resources/stem-engagement-at-nasa/nasa-announces-2025-international-space-apps-challenge-global-winners/ · https://www.nasa.gov/learning-resources/stem-engagement-at-nasa/nasa-international-space-apps-challenge-announces-2024-global-winners/ · https://science.nasa.gov/directorates/smd/2023-nasa-international-space-apps-challenge-announces-10-global-winners/

2.2 The judging score sheet and the evidence we must show

The team's score sheet lists Impact, Creativity, Validity, Relevance and Presentation at 1–20 points each; Teamwork, User Experience and NASA Data Usage at 1–5 each; plus binary checks for naming a 2026 challenge category, public repository access and completing the project page. Criteria 1–5 form the headline 100 points, criteria 6–8 add up to 15 and criteria 9–11 add up to 3, with a 5% women-participation bonus shown separately. The sheet is the team's planning lens, not a guarantee of a score: show evidence for every criterion rather than merely listing features, and verify the latest official participant and judging guidance before submission in case event materials change.

Criterion	What a high-scoring submission shows	Evidence to prepare	Avoid
Impact (1–20)	A meaningful, clearly framed STEM-learning gap addressed through an outpost simulator usable by young learners and educators, with a defined target audience and a plausible path to use beyond one demo.	A concise problem statement, target age range, a learning outcome, 3–5 user tests or observations if possible, and a realistic classroom, club or community pilot plan.	Claiming learning outcomes without evidence, or unsupported global-impact claims.
Creativity (1–20)	A distinctive combination of outpost trade-offs, mission drama, transparent trust mechanics, training mini-games and source-linked science. A replayable loop with crew clues and event twists. Reels are a future expansion, not the novelty claim.	A live playable scenario, a visible consequence system, a moment where one decision creates two linked consequences (not just an animated dashboard), and one memorable narrative moment.	Adding AI or 3D just to sound advanced.
Validity (1–20)	Scientifically plausible, consistent resource rules, sourced facts with correct units, transparent and traceable assumptions, a science review, and a prototype that works as described. Science affects play.	Source citations beside the relevant mechanics, a rule sheet and "model assumptions" note, test scenarios, at least one replayable verification test, a tested demo path, and no inflated AI or sensor claims.	Uncited values described as NASA measurements.

Criterion	What a high-scoring submission shows	Evidence to prepare	Avoid
Relevance (1–20)	Direct mapping to Build a Junior Astronaut Mission Trainer, with lunar or Martian outpost management at the centre. The challenge shapes the core mechanic and deliverable.	The challenge title and problem statement on the project page, README, opening slide and demo narration; each core mechanic mapped to the challenge wording.	A generic space game with a weak challenge connection.
Presentation (1–20)	A concise mission story (problem, data, decision, result, impact) in accessible language, a legible interface and a short, stable live demonstration with a meaningful result.	A rehearsed three-minute demo with a one-minute demo route, a backup recording, captions, short video, clear screenshots and a clear beginning, middle and end.	A feature list with no working demonstration.
Teamwork (1–5)	Visible contribution from research, coding, design, science validation, storytelling and testing.	Named workstream owners, a simple contribution log and repository history showing collaborative work; credit individual contributions.	—
User experience (1–5)	A first-time user starts quickly, understands resources, recovers from mistakes and completes one mission without training.	A one-tap Quick Run, readable interface, short onboarding, keyboard and touch support, a tested fallback, and a five-minute test with someone outside the team that records where they hesitate.	—
NASA data usage (1–5)	Relevant NASA open data or official educational and technical sources are central to a visible feature, with source and attribution clear.	A source register and in-app source panel, a README table listing source, use, unit, date/access and any transformation, and a short explanation of how data affects gameplay.	—
Challenge category identified (0/1)	One exact official 2026 challenge is named.	Use the official title exactly in all materials: Build a Junior Astronaut Mission Trainer.	—
Repository access (0/1)	A working public GitHub or Drive repository with readable instructions and no password barrier.	Test the link in a private/incognito window; include setup and run instructions.	—
Project page complete (0/1)	The NASA project page is filled in and submitted correctly by the official deadline.	One owner and a checklist for every required field: description, challenge selection, data, links, team, media and demo.	—
Women participation bonus (+5% on the sheet)	Meaningful, genuine and documented participation and accurate team representation. Representation is not a substitute for contribution.	Confirm the current official rule, invite inclusive participation, and record roles honestly; never add a name without genuine contribution.	—

NASA open data should be purposeful. For the MVP, use a small number of authoritative resources that genuinely support a mechanic or visual: for example, NASA material on environmental control and life support, space radiation, lunar/Mars environments, and power systems. If using a static value or simplified model rather than live data, label it honestly. Do not claim “real-time NASA data” unless the prototype actually fetches and uses it.

3. Product Concept and Gameplay

3.1 The product at a glance

Item	Definition
Product	CrewReady: Beyond Us
First mission	Mission 01: Dust in the Sunlight (survive the Mars dust storm)
Genre	Cooperative space-outpost survival and systems-management game
Audience	Young explorers roughly ages 10–16; accessible to families and classrooms
Platform	Responsive browser app for phones and laptops
Quick session	3–8 minutes (a short spacecraft voyage with real engineering obstacles)
Story mission	6–10 minutes
Differentiator	Choices affect connected systems; each important decision gets a clear science debrief.
Brand line	A small step towards a universe of possibilities.
Experience promise	One tap to play. A universe of missions to master.

3.2 Two connected loops and the one-tap home screen

Loop	Player experience	Purpose
Instant loop: Quick Run	Tap Play → short spacecraft voyage → 2–4 decisions → outcome → reward → Retry.	Immediate value, low friction and repeat play.
Progression loop: Outpost Campaign	Choose mission → build → operate → react → debrief → earn rewards → upgrade → unlock next mission. Cargo delivered by Quick Run unlocks outpost modules.	Longer-term attachment and progressive learning.

- **Primary button:** PLAY NOW. Opens a Quick Run with the last mode or a safe default.
- **Secondary entry:** CREW UP. Create or join a private room when multiplayer is available.
- **Persistent progression:** OUTPOST. View base, upgrades, discoveries and mission board.
- **Optional cards:** Daily Challenge, Continue Mission and New Discovery. Never block Play Now.
- First meaningful interaction should happen within 15–25 seconds, including a skippable two-step tutorial.

3.3 Quick Run: a spacecraft mission with real engineering obstacles

Quick Run is the one-tap entry point. It is a short survival session with an endless score-and-survival structure, inspired by arcade runners and quick training modes, but it is not an arcade runner and it is not a ten-day real-time wait. It is a real mission: it uses the same science rules and resource engine as the campaign, and it shortens the duration and limits the number of decisions, not the logic. Every run has a clear objective, an explicit starting inventory, at least one coupled systems trade-off, a meaningful consequence and a debrief.

The premise: crew transit to the outpost

The player operates a small crewed spacecraft carrying the first crew and cargo to the Mars outpost site. A spacecraft is a miniature outpost: it must hold power, breathable air, water and food, and protect the crew from radiation, using the same four systems the outpost uses. Each leg of the voyage presents one real engineering obstacle, the evidence about it and two or three solutions, and every solution costs something. Surviving the voyage delivers cargo that unlocks modules in the outpost.

How a run works

1. Launch with one tap. Start with a stable craft, an explicit starting inventory and a clear immediate threat on the telemetry panel.
2. **Every 30–60 seconds a new leg begins:** detect the alert, diagnose the evidence (verifying a sensor if needed), decide by choosing a solution card with a visible cost preview, resolve through the simulation, then learn from one science fact with its source.
3. Obstacles come from a curated pool in a seeded order, so runs vary but remain testable and fair. Difficulty rises through bounded events such as radiation spikes, equipment faults, communication loss, resource squeeze or a sensor warning.
4. At each waypoint (every five legs) the player can bank the cargo delivered so far or continue for a higher score and greater risk.
5. The voyage ends at an arrival gate where the player chooses how to land at the outpost site and hands the cargo to the outpost.
6. After failure, show one reason, one science fact and a one-tap retry. After success, offer the next transfer window with a new seed.

The spacecraft view

A 2D cutaway shows the solar arrays, battery and power bus, habitat cabin, life-support rack, storm-shelter stowage, radiator, hull shield layers and antenna. Tapping a module opens its telemetry. Health is shown in cyan, caution in gold and critical in coral, always paired with an icon and a label. Every essential choice works by tapping.

Scientific obstacles and the solutions players can choose

Each obstacle is based on a real spaceflight problem. All numbers are game-model units, not flight values. Every solution has a cost and no obstacle can be solved for free. Every obstacle offers at least one clue the player can investigate, and each debrief links to a NASA source. Items marked verify must be checked against NASA sources before they are shown as fact.

Obstacle	Real science behind it	What the player sees	Solutions and their costs	Systems affected
1. Solar particle event	Energetic particles from the Sun raise radiation exposure once a craft is beyond Earth's protective magnetic field. NASA's Orion plan uses an onboard radiation sensor to warn the crew, who build a shelter from dense stowage bags of supplies, food and water within about an hour and may stay inside for up to 24 hours.	Dose-rate spike, a forecast with a confidence level and a warning countdown.	A) Build a shelter from stowed supplies: strong protection, costs crew time and blocks access to stores for hours. B) Shelter in the least exposed area and switch off non-essential loads: weaker protection, costs science time. C) Keep working: no cost now, high exposure risk later.	Radiation shielding, crew time, food and water stores

Obstacle	Real science behind it	What the player sees	Solutions and their costs	Systems affected
2. Micrometeoroid or debris strike	Tiny particles travel extremely fast and can puncture a hull. Spacecraft use layered outer shields to break up impacts, and crews patch leaks. Verify against NASA debris and ECLSS sources.	A pressure drop and a leak-rate trend in one module.	A) Seal with a patch kit: uses a spare part and crew time. B) Close the hatch and isolate the module: stops the leak but loses that module's function. C) Add air from the reserve to buy time: spends oxygen.	Life support, spare parts, power
3. Carbon dioxide build-up	Crew exhale carbon dioxide. Scrubbers have limited capacity before they must be regenerated or replaced, and air revitalisation needs working equipment and energy.	CO2 level rising toward a safe-limit marker.	A) Switch to the second scrubber bed or regenerate the first: costs power and a time window. B) Replace the filter: uses a scarce spare. C) Cut crew workload and add ventilation: slower, costs power and delays tasks.	Life support, power, crew time, spare parts
4. Power shortfall	Solar power depends on sunlight, array angle and array condition. Stored battery energy is finite and every system you switch on draws power.	Solar output against load, battery reserve and hours remaining.	A) Re-point the arrays: costs crew time and carries a small fault risk. B) Shed non-essential loads such as greenhouse lights or exercise equipment: saves energy, delays food and fitness. C) Draw down the battery: no immediate loss, less reserve for later legs.	Power, food, crew health
5. Heat build-up	Equipment and crew produce heat, which must be rejected through radiators. Higher power loads create more heat. Simplified for gameplay; verify.	Cabin and equipment temperature trends.	A) Reduce loads: less heat, delayed tasks. B) Reorient radiators: costs crew time. C) Accept a warmer cabin for a while: lower crew performance.	Power, habitat, crew time
6. Communication delay and conflicting information	One-way signals between Earth and Mars take roughly 3 to 22 minutes depending on planet positions (about 1.3 seconds to the Moon), so Mission Control cannot decide for the crew in real time.	A late Mission Control message that disagrees with an onboard sensor.	A) Verify with an onboard diagnostic: costs time and a check. B) Act now with a reversible precaution: costs resources but keeps options open. C) Wait for confirmation: risks an escalating problem.	All systems; evidence and trust
7. Crew health and exercise	In weightlessness, muscles and bones weaken without regular exercise, so crews schedule daily exercise. Verify against NASA Human Research Program material.	Crew fitness and fatigue indicators.	A) Schedule exercise: uses power and crew time and raises air and water use. B) Skip it to save resources: gains supplies now, lowers readiness for landing. C) Shorten the session: partial benefit, partial saving.	Power, life support, crew time

Obstacle	Real science behind it	What the player sees	Solutions and their costs	Systems affected
8. Food and water rationing	Supplies on board are finite and must last the whole voyage plus a margin. Water is recycled, but recovery needs power and working equipment.	Ration and water counts against days remaining.	A) Ration: saves supplies, lowers crew performance. B) Open the reserve: restores performance, shrinks the safety margin. C) Repair the recycler: costs a spare part and power.	Food, life support, power
9. Arrival gate: dust storm at the landing site	Dust storms on Mars can darken the sky and cut solar power, so a crew that lands in one starts with a battery and little sunlight.	A landing-site forecast and a consumables countdown.	A) Land now: cargo arrives, but the outpost starts with low solar output. B) Wait in orbit: protects the outpost start, costs days of supplies. C) Divert to an alternate site: costs extra travel time and supplies.	Power, life support, supplies

Sources to cite: NASA Orion radiation protection plan (https://www.nasa.gov/missions/artemis/orion/scientists-and-engineers-evaluate-orion-radiation-protection-plan/), NASA ECLSS, NASA Space Radiation, NASA Space Place power page, and NASA and ESA material on Earth–Mars communication delay.

Scoring, endless structure and why players return

- **Score:** legs survived, with a safety multiplier for crew health and spare margin. The best score is saved locally.
- **Cargo:** every banked waypoint turns into outpost modules (batteries, filters, shielding stores), so each voyage improves the outpost and gives a reason to return.
- **Daily Voyage:** one shared seed for everyone, with no penalty for missing a day.
- **Fairness:** a curated event pool, bounded difficulty and no unwinnable combinations. A new run always shows what the player could test next time.
- **Destinations:** Mars first (3–22 minute signal delay and dust at arrival). A Moon pack later (about 1.3 second delay and the lunar night) uses the same engine with a new obstacle pool.

How Quick Run answers the challenge

Challenge system	In the spacecraft run	Returns in the outpost
Power	Solar arrays, battery reserve, load shedding	Solar arrays, batteries and dust on panels (Mission 01)
Life support	CO2 scrubbing, leaks, water recovery	Air filter, water recovery and oxygen monitor
Radiation shielding	Solar particle event and shelter built from stowed supplies	Shielding zone and shelter choice (Mission 03)
Food and crew time	Rationing, exercise and workload	Greenhouse, workshop and crew workload (Mission 04)

3.4 Mission 01: Dust in the Sunlight (the Mars outpost)

Premise: a Mars outpost is preparing for a forecast dust storm. Dust can reduce solar-array output; the crew must decide how to protect power generation, preserve life support, and keep the greenhouse within safe operating limits. The mission includes a time budget, limited spare parts, finite battery storage, and a warning signal that may be incomplete or misleading. Players must interpret evidence rather than guess randomly.

Phase	Player action	Science and gameplay rule
1. Assess	Read power, battery, water, oxygen-system status, food stores, forecast and crew condition.	Solar power depends on sunlight and array performance; stored energy is finite. Show units and whether numbers are illustrative.
2. Prioritise	Choose between cleaning/covering an array, moving a rover, reallocating power, or spending a spare component.	Every action costs energy, time, spare parts, or crew effort. Do not allow an action to create free resources.
3. Surprise	A second alert appears: one sensor disagrees with the forecast, or a crew member reports an anomaly.	Present clues and uncertainty. A twist must be explainable by the scenario, not an arbitrary punishment.
4. Decide	Use a limited command window to select a plan and optionally ask another crew role to verify it.	A decision can be correct but costly. Preserve meaningful trade-offs instead of one obvious “best” button.
5. Resolve	Run the simulation and display the cascade across power, thermal control, food production and crew safety.	The event engine calculates outcomes from the same state rules every time for a given scenario seed.
6. Debrief	Explain what happened, what the player missed, and what could be tested on replay.	Teach the causal relationship, identify uncertainty, and distinguish established science from game abstraction.

Gravity and movement must also remain plausible. In transit the crew and loose objects float because the craft is in free fall. On a planetary surface gravity is reduced, not zero: objects do not float away without a force, and inertia, traction, suit mobility and rover handling should be represented in simple, consistent ways. Mars and the Moon have different gravity and atmosphere. The game should never use “zero gravity” for surface operations. If a mechanic is simplified for playability, label it as a simulation abstraction in the debrief.

3.5 The survival model: finite resources, linked systems, no magic fixes

Use a transparent resource ledger. Resources can be gained, consumed, lost, recovered, traded or donated, but every change must have a cause. Resource quantities and rates in the MVP should be clearly marked as game-model values unless they come directly from a cited NASA source. Avoid pretending the game is a certified mission-planning simulator.

System / resource	What the player manages	Realistic dependency to model
Power	Solar output, battery charge, essential and optional loads, spare equipment	Power generation varies with sunlight, array condition and location. Batteries store finite energy; turning on a system draws power.
Life support	Oxygen availability, cabin air quality, water inventory, recycling status	Crew consumes oxygen and produces carbon dioxide; air revitalisation and water recovery require functioning equipment and energy.

System / resource	What the player manages	Realistic dependency to model
Radiation protection	Shelter choice, shielding material, warning time, exposure budget	Radiation risk depends on the event, duration, location and shielding. A shelter reduces exposure; it does not make risk vanish.
Food and agriculture	Rations, stored food, crop growth, water, nutrients, light and crew time	Crops need time and inputs. A greenhouse cannot produce instant food; stored rations bridge the time before harvest.
Thermal / habitat	Temperature, insulation, heat rejection, equipment operating range	Power, sunlight/shadow, habitat design and equipment loads affect thermal control.
Crew health and time	Fatigue, workload, task time, injury/illness events and morale	Crew cannot perform unlimited tasks at once. High workload and delayed maintenance can increase risk.
Spare parts / mass	Filters, seals, cables, repair kits, shielding mass and cargo capacity	Spare inventory is limited. Bringing or installing one item can create a mass, power, time or maintenance trade-off.
Trust / communication	Verified reports, uncertainty, promises, shared inventory and role confidence	Trust is a decision-support variable, not a supernatural resource. It affects cooperation and information-sharing, not the laws of physics.

- **Gain:** recover water through a functioning recycling system, generate electricity under suitable conditions, harvest crops after the required growth period, or receive a scheduled cargo delivery if the scenario includes one.
- **Loss:** consume rations, use power, lose water through a leak, spend a filter, take radiation exposure, or lose time during a repair. Losses must appear in the ledger with a reason.
- **Recover:** a repair or maintenance action may restore a system, but only after paying its time, energy and part costs. Recovery is not guaranteed if the system is damaged beyond the chosen scenario's repair capacity.

3.6 Campaign: ten stages, not ten real-world days

> **RETENTION DECISION**
> A ten-mission campaign means ten short play sessions, not ten days of waiting. Each mission takes roughly 6–10 minutes. Players can binge several stages or return later without losing progress.

Use the feeling of escalating missions in story-driven games, but do not copy any game's criminal themes, interface, characters or missions. Each stage introduces one new systems challenge, then combines previously learned systems in a more demanding scenario. The campaign has mission briefings, objectives, optional objectives, unexpected turns, debriefs, unlockable equipment and a persistent outpost. Progression is earned through understanding and decision quality, not grind or paid advantages.

Stage	New pressure introduced	Example mission arc	Reward / unlock
01. First Signal (First Footprint)	Basic power, batteries, status indicators and the resource ledger.	Learn the outpost, identify essential loads, survive a short power dip.	Basic battery module

Stage	New pressure introduced	Example mission arc	Reward / unlock
02. Thin Reserves	Water and air revitalisation.	A leak forces the crew to prioritise recovery and rationing.	Water-recovery unit
03. Red Sky / Solar Alert	Radiation shelter and warning uncertainty. A warning may be a sensor fault, so verify before spending resources.	Choose a shelter plan while a time-sensitive repair is unfinished; use diagnostics to tell a false alarm from a real anomaly.	Shielding zone; diagnostic scanner
04. Greenhouse Clock (The Quiet Greenhouse)	Food, water, light and time. Food production competes with power and water.	Protect crops without draining energy needed by critical systems.	Greenhouse module
05. Dust in the Sunlight (Dust on the Horizon)	Solar performance and maintenance. Solar generation drops and critical loads must be prioritised.	Forecast uncertainty, dust, spare parts and battery limits collide.	Power-management upgrade
06. The Missing Filter (Spare-Part Problem)	Inventory, investigation and trust. Repair one system with scarce parts or reduce demand.	Trace a mismatch in stock records before choosing a repair plan.	Workshop module; spare-parts rack
07. Distress Channel (Signal Lost)	Donation and shared survival. A communication fault creates uncertainty.	Help another crew without making the home outpost unrecoverable.	Comms relay
08. Two Emergencies (The Crew Disagrees)	Prioritisation under a deadline. Different roles receive different evidence.	Two systems fail close together; a team cannot do every task and the crew disagrees on what to fix first.	Team cosmetic
09. The Long Night (Cold Night)	Storage, thermal control and long-horizon planning. Heating competes with other power needs.	Prepare for a period of limited solar input with finite batteries: a long dust-darkened sky on Mars, or the lunar night in a Moon pack.	Insulation module
10. Beyond Us (Homeward Signal, Storm Peak)	Integrated systems and co-op command. A final systems check combines familiar hazards with one new twist.	A multi-stage crisis tests resource planning, communication and trust; earlier decisions shape the ending.	Advanced blueprint; campaign completion and replay modifier

Developers can keep adding mission packs after the hackathon: new lunar sites, Mars scenarios, solar events, equipment failures, research objectives and classroom-designed challenges. Each pack must pass a science review, have testable rules, and add a new learning objective rather than simply increasing numbers.

3.7 Thrills, twists and fair bluffs

The desired feeling is tense survival and team judgement, not random punishment. Use twists, incomplete information, competing interpretations and costly choices. Any bluff should come from a character's limited knowledge, a simulated sensor fault, a conflicting priority or a strategic teammate decision. Do not present fabricated science as fact.

Twist design principles

- **Foreshadowing:** Nova spots a strange sensor reading before the alarm.
- **Uncertainty and false alarms:** a sensor may be faulty, but clues let players evaluate it; checking diagnostics can reveal a faulty sensor and avoid wasted resources.

- **Hidden dependency:** a greenhouse helps future food but increases immediate power and water demand.
- **Equipment damage:** a system degrades gradually, allowing a response before crisis.
- **Crew dilemma:** save power now or protect future food production; both options have understandable costs.
- **Story twist:** apparent sabotage is a damaged relay or environmental fault, not a frightening human villain.
- Use suspense, alarms and countdowns without gore, graphic death, jump-scare audio or punishment for missed logins.

Twist and event catalogue

Twist / mission event	Decision tension	Fairness rule
Conflicting sensor readings	Spend time verifying, act on the stronger signal, or choose a reversible precaution.	Provide at least one clue that can be investigated.
Solar storm warning window	Move crew to shelter, protect critical systems, or continue an urgent repair.	Explain warning confidence and the cost of delay.
“We have enough power” claim	A teammate or NPC overlooks a hidden load. Players inspect the power budget or trust the report.	The load is discoverable in telemetry; do not trick the player with invisible rules.
Damaged water recovery unit	Spend a rare spare now or ration water while attempting a slower repair.	Both paths have plausible costs and neither is always best.
Greenhouse emergency	Keep crop lighting on or reserve energy for life-support equipment.	Show that food production has time delays and cannot instantly replace rations.
Distress signal from a nearby crew	Donate a limited resource, offer a repair plan, or decline to preserve the home outpost.	Show the donor’s resulting risk and the recipient’s verified need.
False alarm or genuine anomaly	Use a verification action, trust the crew member, or take a precaution.	The answer is seeded and explainable; evidence and uncertainty are visible.
Trust test	A teammate requests an item but has not shared all relevant evidence. Ask questions, verify logs, or transfer a limited stock.	Consequences reflect information and cooperation, not a simplistic good/bad morality score.

3.8 Base building and upgrades

Upgrade path	Example modules	Visible trade-off
Power	Solar array, battery bank, power controller	Build cost versus reserve and energy demand.
Habitat	Insulation, shielding zone, sealed compartment	Supplies now versus protection during an event.
Life support	Water recovery, air filter, oxygen monitor	Maintenance/energy versus resilience.
Food	Greenhouse, seed cabinet, grow-light controls	Power/water now versus future food.
Engineering	Workshop, spare-parts rack, diagnostic scanner	Construction cost versus repair options.

Upgrade path	Example modules	Visible trade-off
Communications	Relay, antenna, backup transmitter	Power/supplies versus information reliability.

Keep build supplies and energy as separate quantities. Each module should have a visible cost and effect. Upgrades must alter gameplay, not only appearance.

3.9 Crew Training Deck: brain and body training as mission readiness

Training is a supporting system connected to mission outcomes and to the outpost simulation, not a separate fitness app. It must never be presented as a diagnostic tool, a validated astronaut assessment or a replacement for professional physical training, and the game must never claim these mini-games medically measure fitness or diagnose a user. A short training task can prepare the player for a later decision, and the mission can show how attention, spatial reasoning, communication and workload management matter to crew operations. The hackathon prototype should implement one brain activity and one safe body-awareness activity, then show how each contributes to a mission task.

Module	Prototype activity	Mission connection	Safety and validity guardrail
Brain: working memory	Remember and repeat a short sequence of warning lights, switches, status alerts or supply codes.	Supports checklist discipline before a repair mission and helps prioritise a control panel during an emergency.	Describe it as a game exercise, not a clinical or cognitive test.
Brain: attention, reaction and anomaly spotting	Respond to a changing visual signal while ignoring irrelevant signals, or find an unusual sensor reading in a control-room panel.	Supports prioritising an alarm among several non-critical indicators and verifying a warning before spending scarce supplies.	Measure only in-game performance; do not claim real-world reaction ability. Reward checking evidence rather than tapping quickly.
Brain: spatial reasoning and orientation	Rotate or align simple modules to connect a habitat, power unit or route, or route a rover around terrain using a simple map.	Helps players understand orientation, system layout, docking and surface navigation.	Keep controls accessible, use consistent coordinate and movement rules, and provide a non-timed mode.
Brain: risk decisions and judgement	Choose between two or three actions, or compare two plans, with different probabilities, costs and uncertainty.	Builds risk awareness and systems thinking: choose whether to repair, shelter, ration or donate.	Explain trade-offs and show uncertainty; avoid implying one simplified scenario represents all real missions.
Body: balance and controlled movement	Optional standing reach, pose-guided movement, seated upper-body mobility task or low-intensity movement sequence.	Links movement to a simulated maintenance task and teaches that movement in a suit is constrained and that tasks consume crew time and energy.	Opt-in only. Use MediaPipe pose detection only if feasible, with camera-off alternatives, no scoring of body shape and no medical or health claims.
Body: coordination sequence	Follow a short on-screen movement sequence or tap pattern.	Represents procedural coordination in a playful training context.	Optional, low impact, accessible and suitable for limited space.
Team communication	One player sees the sensor panel, another sees inventory; they must explain their evidence.	Practice handoffs and mission communication.	Keep roles fair and offer a solo-accessible alternative.

Recommended MVP: one 30–60 second memory/checklist activity and one optional movement or coordination activity. Use browser-friendly input first; add camera-based pose estimation only after consent, device testing, privacy checks and a reliable fallback. If camera-based pose detection is attempted, request permission clearly, process locally where feasible, do not store raw video or upload pose data, and provide a button-based alternative. In the hackathon build, create brain and decision training first; leave pose detection as a stretch goal unless it can be demonstrated reliably.

3.10 Crew play: roles, multiplayer, sharing and trust

The desired experience is up to four players on separate phones, joining a private room with a simple code. Real-time networking, reconnection and state synchronization can consume more time than the game, so multiplayer must be staged.

Tier	Mode	Implementation / decision
MVP fallback	Solo or pass-and-play	Always available; no network dependency. Use to prove the core loop.
Tier 1	Same-Wi-Fi room, up to four phones	Host creates room and shows code/QR. Test a local-network or peer-to-peer approach on target devices; browser LAN discovery is not consistently automatic.
Tier 2	Online room for four phones	Room code, ready state, role assignment, shared state, disconnect handling. Prefer a small authoritative room service or managed realtime backend.
Tier 3	Public matchmaking, friends, voice/chat, rankings	Defer due to moderation, privacy, backend and abuse-prevention burden.

Crew roles

Role	What they see / do	Example clue
Commander	Objective, timer, final decision or vote.	Knows the crew-safety priority.
Flight Engineer	Power draw, battery and equipment status.	Sees which system is consuming too much energy.
Life-Support Specialist	Oxygen, water, air and habitat status.	Knows which resource cannot be cut safely.
Scientist / Mission Control	Weather warning and sensor confidence.	Can distinguish a likely hazard from a faulty reading.

Solo play must expose all critical information through role tabs. Multiplayer adds communication and teamwork, not an impossible information gate.

Networking realities

- A normal web page cannot reliably discover all devices on the same Wi-Fi without browser and network constraints. Test the actual connection approach on the target phones.
- For a hackathon demo, a four-phone online room may be more reliable than automatic LAN discovery. Keep solo mode as fallback.
- Avoid voice and public text chat for children. Use preset messages such as “Check power”, “Verify sensor” and “Protect life support”.
- If networking fails, the game must still run and the demo must have a reliable solo route.

Sharing resources and building trust

The cooperative premise should make helping another crew a real strategic choice. Each outpost has a finite inventory, a limited power budget, crew time and repair capacity. A request for help should show the requested quantity, verified need, transfer cost, delivery delay if relevant, and the donor's resulting reserve. The recipient can accept, counter-offer, share evidence or propose a reciprocal task. Helping is meaningful precisely because it is not free.

Mechanic	Rule
Shared stock	A transfer subtracts the same amount from the sender and adds it to the receiver, with a transaction log and a visible receipt. No duplication, unlimited donation or negative stock. In co-op, a player can allocate a limited shared stock item or power budget to another station or crew role.
Power sharing	Represent as a scenario-specific allocation or delivery system, not instant magical energy transfer between distant habitats.
Trust	Built through accurate reports, fulfilled commitments and transparent resource logs. Teammates can share evidence, flag uncertainty, ask for a second opinion or commit to protect a shared reserve. Concealment or a broken promise reduces a team-trust score and may reduce cooperation or information reliability in later scenarios, but never secretly changes a scientific outcome.
Bluff / uncertainty	NPCs or players may be uncertain or strategic, but the game distinguishes deliberate deception from sensor uncertainty. Clues can be checked.
Failure and recovery	A crew may fail a mission yet preserve learning, unlock a debrief and retry with new information. Do not erase all progress arbitrarily.
Accessibility	Solo mode must simulate crew roles so a player is never blocked from the core learning objective by a lack of friends or internet.

4. MVP, Feature Tiers and Priorities

The wider vision is large and the MVP is narrow. Tiers describe where each feature belongs in the product roadmap. Priorities (P0 to P3) describe what the team builds first and when a feature counts as done. P0 matches Tier 0, P1 matches Tier 1, P2 covers stretch work from Tier 2 and Tier 3 concepts, and P3 is post-hackathon work from Tier 3 and Tier 4.

4.1 Feature tiers

Tier	Features	Delivery gate
TIER 0 · MVP / competition core	Responsive browser app; one-tap Quick Run; one complete Mars outpost mission; setup/build choices; separate supply and energy; 1–2 events; visible consequences; outcome; science debrief and sources; retry; local save; polished home screen; reliable demo path.	Mandatory. No submission without a complete end-to-end playable mission.
TIER 1 · Strong demo extension	Four crew roles; asymmetric clues; private room for four phones if feasible; QR/room code; preset messages; one campaign chain; basic upgrades; curated Quick Run event pool; daily seeded challenge.	Only after Tier 0 is stable.
TIER 2 · Retention and depth	Ten-stage campaign; upgrade trees; achievements and Space Passport; mission modifiers; weekly challenge; avatar cosmetics; Discover Library; Earth Detective using credited satellite imagery; short experiments.	Post-MVP or after a stable core.
TIER 3 · Community content	Landscape Space Reels; creator submissions; captions; categories; approved creator profiles; attribution; editorial queue; educator playlists; creator reward programme only after legal and safety review.	Later. Never launch an unmoderated feed.
TIER 4 · Long-term platform	Online accounts and sync; public matchmaking; classroom dashboard; Moon/Mars mission packs; co-op events; optional AI-generated mission variations with human-reviewed science; offline content packs.	Only after user testing and operational capacity.

Optional features that must earn their place

- **Brain training:** one short memory/reaction activity can later fit a training room if it directly connects to mission readiness.
- **Body training and camera pose detection:** defer because of permissions, privacy, accessibility and device variability.
- **Earth Detective:** add only with a meaningful Earth-observation dataset and a clear challenge fit.
- **AI:** use only for a defined benefit, such as explaining a debrief with source-grounded content. The game must work without AI.
- **Creator earnings:** defer until moderation, rights, funding, and lawful safeguarding processes are ready. Never promise children direct earnings in the MVP.

4.2 Build priorities and definition of done

Priority	What to build	Definition of done	Do not build yet
P0	Resource simulation engine	Power, life support, water, food, radiation exposure, crew time and inventory update deterministically with clear units or labelled abstractions.	Large campaign, open multiplayer, user accounts, AI chatbot, creator monetisation.
P0	Quick Run: the spacecraft voyage	One-tap start; at least five science-reasoned obstacles; every solution has a cost; a source-linked debrief; seeded events; instant retry.	—
P0	One complete Mars outpost mission (Mission 01)	Player assesses, chooses, sees consequences, can fail or succeed, and receives a science debrief.	—
P0	Readable mobile-first control room	Start mission, inspect resources, choose actions and understand warnings on a phone screen.	—
P0	Source and assumption panel	Each science claim or value has a source or is marked as a game-model assumption.	—
P0	Challenge, data and submission basics	Official challenge named; one visible NASA data or resource link; three meaningful decisions; consequence feedback; end-of-mission explanation; public repository; completed project page.	—
P1	Mission progression and twists	At least three distinct scenario variations, each with evidence, choices and explainable outcomes.	Advanced pose scoring, elaborate progression economy, broad device-support promises.
P1	Co-op resource transfer prototype	Shared inventory, logged transfers, limited stock and a clear solo fallback.	—
P1	Brain training activity and crew roles	One memory, spatial or anomaly-spotting exercise that connects to a mission mechanic, plus simple crew roles, captions, a source panel and a responsive layout.	—
P2	Body / pose training	Only if safe, permissioned, optional, tested and achievable after P0 and P1.	—
P2	Space Reels concept	Future-feature mock-up and roadmap only; it must not displace the mission simulator.	—
P3	Online rooms, creator rewards, large campaign, classroom dashboards, more mission packs, curated creator marketplace, educator partnerships, multilingual packs, deeper privacy-protecting analytics	Post-hackathon roadmap; needs security, moderation, hosting, accessibility and content governance.	Public creator payouts before safety, rights, moderation and legal processes exist; an unmoderated social network; infinite-scroll incentives; unsupported “AI astronaut readiness” claims.

5. Product Requirements and Design

5.1 Product Requirements Document (PRD)

Goals

- Teach connected space-system trade-offs through play.
- Let a new user start in one tap with minimal reading.
- Create a satisfying retry after both success and failure.
- Make scientific claims traceable and distinguish facts from game estimates.
- Demonstrate a stable, complete product within the hackathon scope.

Non-goals for the MVP

- No public social feed, public chat, direct messaging or user-uploaded videos.
- No required account, real name, exact birth date, precise location, face photo or camera access.
- No full 3D world, complex physics engine or huge open world.
- No monetisation, loot boxes, pay-to-win upgrades or child-targeted ads.
- No claim of flight-certified engineering accuracy or real mission forecasting.

User stories and acceptance criteria

User story	Acceptance criteria
Start immediately	Play Now begins a run in at most two taps; tutorial is skippable; first meaningful choice arrives quickly.
Feel decisions matter	Each action changes at least two state values, or one value plus a future option; debrief explains why.
Want to replay	Quick Run uses a curated event pool with bounded difficulty and no unwinnable random combinations.
Improve the outpost	Upgrades show cost and effect before purchase; effect appears in gameplay.
Help a crew	Each role has a useful clue; clues can be shared through safe preset messages.
Know what is real	Major science cards have sources; fictional values are labelled game estimates.
Stay safe	No public messaging, personal data collection, public profiles or unmoderated uploads in MVP.
Judge can verify it	Demo works, main path completes, source links work and challenge title is explicit.

Prototype success metrics

Metric	Initial target	Test
Time to first play	≤25 seconds	Observe a new user from landing screen.
Mission clarity	Most testers finish without coaching	Test with 3–5 people unfamiliar with the app.
Decision understanding	Player explains one trade-off afterward	Ask what changed and why.

Metric	Initial target	Test
Retry intent	Player voluntarily retries or asks what else can happen	Observe behavior, not just survey answers.
Science transparency	Player can distinguish source facts from game values	Ask which numbers are real data.
Reliability	No blocker on primary device; backup recording ready	Repeat demo path at least five times.

5.2 UI/UX design system

Visual direction

Make children feel they entered a lively mission-control room, not a school portal. Combine friendly astronaut characters, short labels, game-like progress and a calm futuristic interface. Use glassmorphism selectively for featured overlays, not behind every paragraph.

The art direction is “deep-space sky”: cinematic dark navy for focus, luminous cyan for live systems, nebula violet for navigation, warm gold for decision prompts, and coral only for urgent danger. Use the colours consistently so a child can learn what each signal means. Do not rely on colour alone: pair it with labels, icons and patterns.

Token	Hex	Use
Deep Space Navy	#10172F	Main background, control room, navigation, night-sky surfaces and high-contrast panels.
Nebula Violet	#6C63FF	Primary actions, active navigation, selected states, mission progression and highlights.
Cosmic Cyan	#51E5E0	Healthy and live system status, active states, data traces and oxygen, water and power accents.
Star Gold	#FFCB69	Rewards, discoveries, choices, key objectives, resource gains and caution that is not yet critical.
Rocket Coral	#FF7C9C	Critical warnings, urgent events and danger. Reserve it for high priority and keep the tone non-punitive. Character art may echo it, UI signals may not.
Moon White	#F5F7FF	Main text and bright surfaces on dark backgrounds.
Orbit Blue	#78BDF2	Secondary data, links, map paths and informational visuals.
Muted Starlight	#AAB7D4	Secondary labels, timestamps and less prominent telemetry.

UI treatment: rounded but not childish cards, soft glow used sparingly, layered star-map backgrounds, clear resource bars, animated cause-and-effect, and reduced-motion support. Keep the mission control screen calm enough to scan during a crisis. Avoid excessive glassmorphism over small text; readability and touch usability take priority over decoration.

Interaction principles

- One dominant action per screen; Play Now stays easy to find.
- Large tap targets, readable contrast, short labels and no tiny dashboard text.
- Rounded cards, visible hierarchy, icons plus labels and selective glass panels.
- Short purposeful animations; provide reduced-motion and mute controls.

- Sound is optional; no jump-scare audio.
- Use original or correctly credited/licensed imagery.
- Include loading, empty, success, pause, reconnect and error states.
- Never communicate danger by colour alone.
- All essential choices work by tapping, not only dragging.

Screen-by-screen blueprint

Screen	Content and interaction	UX requirement
1. Launch / Home	CrewReady: Beyond Us; Nova; PLAY NOW; CREW UP; OUTPOST; mission progress; Daily Voyage card.	Instant entry. No forced account or long onboarding.
2. Quick Run briefing	One-sentence goal, threat, two system readings, Launch.	Teach only what is needed now.
3. Astronaut / role	Choose avatar colour or role; skip available.	Customisation takes seconds.
4. Outpost setup	Choose modules; show supply cost, energy demand, benefit and drawback; base illustration updates.	Preview consequences before confirm.
5. Mission Control / spacecraft view	2D cutaway or isometric base; power, life support, food, shielding, crew safety; current phase and alerts.	Show the most urgent system first.
6. Event / crisis	Nova introduces the incident; evidence and clues; 2–3 solution cards with a short cost preview.	Suspense without confusing text.
7. Crew room	Room code/QR, ready states, roles, connection and preset messages.	Clear join, leave and reconnect; no public chat.
8. Outcome	Mission status, state changes, best decision and biggest risk.	Explain why, not only win or lose.
9. Science debrief	Short explanation, source link, simplification note, Try Again.	Source is one tap away.
10. Outpost / upgrades	Upgrade branches, cost and effect preview, unlocked modules, cargo from Quick Run and mission board.	Every upgrade changes gameplay.
11. Space Passport	Mission badges, discoveries and progress.	No streak shame or penalty for absence.
12. Discover / future content	Curated science cards and later creator content if moderated.	Never ship a blank library or an unmoderated feed.

Onboarding with Nova

1. **Nova:** “Welcome, explorer. Your crew’s outpost needs power to stay safe.”
2. Show one resource bar and one alert. Ask the player to choose between two simple actions.
3. Animate the state change and explain it in one sentence.
4. Offer Play Now. Let the player skip or replay the tutorial at any time.

UI references and design logic

Borrow interaction patterns, not copyrighted art, names, sounds or assets. The interface should feel instantly understandable to children and still look credible to an older student or adult judge. Use a

mobile-first, responsive web interface, large touch targets, strong visual hierarchy, and a clear distinction between the fictional mission narrative and real scientific explanation.

Reference	What to learn from it	How CrewReady adapts it
Among Us	Simple crew roles, readable status, tension from incomplete information and discussion.	Crew roles and anomaly checks, but no social deduction that overwhelms science; evidence and rules remain inspectable.
Minecraft	Learn-by-doing, visible systems, building and persistent world progress.	Outpost upgrades and resource consequences; avoid unbounded crafting that makes supplies feel infinite.
Roblox / accessible game hubs	Clear entry points, avatar identity, varied missions and creator participation.	Mission selection and future creator challenge submissions with moderation; not an open unmoderated world.
GTA mission progression	Clear objectives, mission briefings, escalating stakes, unlocks and a sense of campaign.	A non-criminal, science-led mission arc with optional objectives and meaningful debriefs.
YouTube Shorts / TikTok vertical feeds	One-thumb horizontal/vertical discovery, short visual hooks and low-friction next-item browsing.	A future swipeable “Space Reels” feed of 15–60 second science explainers, data visuals and creator experiments. Never interrupt an active emergency mission.
Duolingo	Short sessions, progress cues, streak-like continuity and immediate feedback.	Optional mission readiness goals and learning milestones; no shame, manipulative streak loss or pay-to-win.
NASA Eyes / NASA data visualisations	Scientific context, visual exploration and connection to authentic data.	A “show the data” layer, sourced visualisations, uncertainty labels and links to NASA references.

6. Technical Requirements

6.1 Recommended architecture

Use a responsive web stack the team can already debug and deploy. Separate simulation rules from UI so the game can be balanced without rewriting screens. A stable local scenario is more valuable than a fragile live-data dependency.

Layer	Recommendation	Reason
Client	React / Next.js or another familiar browser stack; mobile-first CSS.	Fast iteration, responsive UI and easy demo.
Game state	Deterministic simulation module with documented state.	Testable rules and explainable outcomes.
Scenario config	JSON/TypeScript config for resources, upgrades, event triggers and effects.	Tune values without rewriting core logic.
Persistence	Local storage for MVP progress and settings only.	No login or backend needed.
Science/data	Curated NASA source links and cached, timestamped data where appropriate.	Demo still works if an API fails.
Multiplayer	Room state service or managed realtime backend; authoritative shared state.	Avoid conflicting phone states and support reconnect.
Hosting	Reliable HTTPS deployment plus a local fallback.	Easy testing and judging.
Analytics	Manual usability observation in MVP.	Avoid unnecessary tracking of children.

6.2 Game-state model and simulation rules

State group	Example variables	Rule
Resources	buildSupplies, batteryEnergy, solarOutput, waterReserve, foodReserve	Separate units; define min/max and update timing.
Systems	lifeSupportStatus, equipmentCondition, shieldingLevel, greenhouseOutput	Every module has explicit costs and effects.
Crew	crewSafety, workload, missionConfidence	Fictional game indicators, not medical predictions.
Scenario	missionId, phase, eventSeed, eventHistory	Reproducible runs for debugging and demo.
Progress	completedMissions, unlockedModules, earnedBadges	Local-only in MVP; no public profile.
Room	roomCode, temporaryPlayerId, role, ready, stateVersion	No personal identity; expire room data after session.

- Every decision affects at least two understandable variables, or one variable plus a future option.
- Offer several viable strategies; do not require guessing hidden rules.
- Each event has a trigger, evidence, options, state effects and a plain-language explanation.
- Bound values and handle shortages explicitly.

- Keep balance values in one editable configuration area.
- Test all outcome paths and ensure UI matches the underlying state.
- Label simplified values as game units or estimates, never actual mission thresholds.
- Quick Run and the campaign call the same simulation step function. Quick Run changes only the scenario configuration, the obstacle pool and the clock.
- Each obstacle or event card stores an id, trigger, evidence shown, two or three solutions with costs, state effects, a plain-language explanation, a science fact and a source id.

6.3 NASA data plan and scientific integrity checklist

Implementation requirement: keep a source register with resource title, exact URL, access date, data fields used, units, transformations, limitations, and the specific mechanic or visual it informs. In the interface, distinguish NASA-derived information, other sourced facts, and fictional balancing values.

- NASA Moon to Mars / mission resources and Planetary Data System: use for future lunar or Mars terrain, environment, and mission context after the MVP is stable. Avoid introducing a dataset merely as decoration.
- **NASA Space Crops resources:** use to ground food-production trade-offs such as water, nutrients, lighting, crop health, time-to-harvest, and energy demand. Do not imply that a greenhouse instantly produces food or has no resource cost.
- **NASA Open Science Data Repository (OSDR):** a possible future source for spaceflight biological, physiological, behavioural, plant, and environmental research. Use only if the team can connect a specific dataset to a clear, age-appropriate game decision.
- NASA Open Data Portal (data.nasa.gov) and NASA Science Data Portal: use these to discover relevant datasets. Record the actual dataset identifier and follow the link to its authoritative archive; the portal may provide metadata rather than host the data itself.
- **NASA Open APIs (api.nasa.gov):** investigate suitable APIs for space-weather events or NASA imagery where they add a real mechanic. Check current endpoint documentation, rate limits, API-key requirements, and whether the chosen endpoint remains supported before integrating it.
- **NASA Space Place, “What Powers a Spacecraft?”:** use as an accessible learning reference for solar power and energy systems. For the dust mission, label any reduction in panel output as a scenario assumption unless it is tied to a cited dataset or documented physical model.
- **NASA Human Research Program, Space Radiation:** use to explain radiation as a real crew hazard and to ground shelter/alert scenarios. Distinguish radiation exposure concepts from fictional shield-strength points or simplified gameplay thresholds.
- NASA Environmental Control and Life Support Systems (ECLSS): use for the real functions and dependencies of air revitalisation, oxygen, water recovery, waste management, and environmental monitoring. Translate these into simplified, labelled game mechanics, not a claim that the game reproduces flight hardware.

Use only sources that directly affect a mission rule, event, visual, or learning debrief. The sources below are candidate inputs to verify and document during implementation; listing a source does not mean its raw data are already integrated into the prototype.

Scientific integrity checklist

- Choose the official 2026 challenge first, then select data/resources that directly support its problem. Do not choose a dataset only because it looks impressive.
- Make at least one data-informed output visible in the product: a chart, map, event, parameter, scenario input, or evidence card.
- Record the dataset/resource name, provider, URL, retrieval date, relevant fields, transformation, and limitations.
- Distinguish observed/source values from simplified game mechanics. If the game uses illustrative values, label them clearly.
- Cite NASA and third-party sources in the UI and README. Check current event rules for attribution, AI disclosure, media labelling, and use of NASA branding.
- Test the actual project link and repository access on a device or browser that is not logged into the team account.

6.4 Offline, privacy and child safety

- Solo Quick Run should work after the app shell and scenario assets are cached.
- Show offline, reconnecting and shared-room states clearly.
- If multiplayer drops, pause briefly, offer reconnect, and let the host continue or safely end the room.
- No public chat, direct messages, public profiles or camera requirements in MVP.
- Use temporary room identities; do not collect real name, exact birth date or location.
- Later uploads require moderation, reporting, copyright checks and appropriate consent/safeguarding review.
- No creator payment promises to children; any future rewards need legal and operational review.
- Use captions, readable contrast, reduced motion and sound-off play.

7. Build Plan: Steps, Tasks and Roles

7.1 Ground rules before building

The hackathon is time-boxed. Confirm with the Local Lead what preparation, existing concepts, code and assets are allowed before the event. Follow the current official rules rather than assuming all pre-event work is either permitted or prohibited.

The 2026 Team Formation Guide says participants should not begin working on their project until the hackathon begins on 14 November 2026 at 9:00 a.m. local time. Registration, team formation, reading the challenge, planning and practising tools on unrelated projects are the safe preparation. The game, art, final copy, videos and project page are built from the official start. Confirm the details with the Local Lead.

7.2 Step-by-step build order

Order	Build	Why
1	Quick Run spacecraft voyage with three obstacles and one visible consequence, then grow the pool to five	Proves the core mechanic.
2	Science debrief, source labels and retry	Makes it a science-learning experience.
3	Outpost setup and one meaningful upgrade	Adds strategy.
4	Polish home screen and mission visuals	Improves first impression after logic works.
5	Multiplayer room spike if staffing/time allows	Adds teamwork without risking the core demo.
6	Extra missions, reels, accounts or AI	Only after the above is stable.

7.3 Workstreams, tasks and definition of done

Workstream	Tasks and subtasks	Definition of done
A. Challenge and scope	Read official challenges; select best fit; map requirements to mechanics; define audience; lock MVP; confirm rules.	One-page challenge map and agreed scope.
B. Science/data	Research NASA life support, Mars dust/power, radiation and food systems; select a sourced element; record URL, date, units and limits; write source drawer.	Every claim has a source or game-estimate label.
C. Game design	Write objective; define state/resource rules; design 1–2 crises; define win/fail/retry; draw six-screen flow; define role clues.	Complete game-design sheet and edge cases.
D. UI/UX	Build home, briefing, setup, dashboard, event, outcome/debrief; apply tokens; create loading/error/reconnect states.	New user finds Play Now and understands a decision.
E. Simulation	Implement state model, costs, upgrade effects, event engine, outcomes, retry reset and local progress.	A complete run works without manual state editing.
F. Multiplayer spike	Test room on target phones; room code/QR; role assignment; ready states; shared decisions; disconnect handling; solo fallback.	Four-device demo only if stable; otherwise clearly defer.

Workstream	Tasks and subtasks	Definition of done
G. QA/science	Test all paths, repeated runs, small screens, contrast and tap sizes; validate source links; test with 3–5 users.	No mission-blocking bugs; users can explain one trade-off.
H. Pitch/media	Write video; label mockups; record demo; build 3-minute pitch; prepare backup; rehearse judge questions.	Working demo, concise pitch and backup recording.
I. Submission	Complete official fields, challenge response, description, links, credits and data sources; test public access.	Submission confirmed before deadline.

7.4 Team roles

Role	Owns	Coordinates with
Product lead	Scope, challenge mapping, priorities, task board.	Everyone
Developer	Simulation, integration, deployment.	UI/UX, science, QA
Science/data lead	Sources, claims, assumptions, debrief.	Developer, pitch
UI/UX designer	Screen flow, visual system, assets.	Product, developer
Story/pitch lead	Video, demo script, submission narrative.	Science, product
QA/integration	Test plan, devices, links, backup demo.	Everyone

In a small team, one person can hold multiple roles. Each task still needs one owner, a deadline and a definition of done.

7.5 Suggested weekend schedule

The hackathon runs from Saturday 9:00 a.m. to Sunday 11:59 p.m. local time, about 39 hours. Replace these times with the Local Lead's timetable once it is confirmed.

When	Goal	Done when
Sat 9:00–10:00	Read the final challenge text, lock scope, set up the repository, deployment and shared state types, assign tasks.	A blank page is live at a public URL.
Sat 10:00–17:00	Simulation engine with tests. Quick Run spacecraft screen on the engine with three obstacles.	A player can tap Play, lose, read the debrief and retry.
Sat 17:00–22:00	Mission 01 outpost, upgrades, source panel, and the Quick Run pool grown to five obstacles.	An upgrade changes the next run.
Sat 22:00–Sun 5:00	Sleep in shifts. One person watches the live link.	A rested team.
Sun 5:00–12:00	Daily Voyage, data labels, language and accessibility settings, crew roles with a solo fallback, one brain mini-game.	A judge can play well with no help.
Sun 12:00–15:00	Playtests with 3–5 testers, stretch items only if stable.	Feature freeze at 15:00.
Sun 15:00–18:00	README, source register, videos, project page.	Submitted by 18:00.
Sun 18:00–23:59	Buffer: fix problems and rehearse the pitch.	Nothing new is added.

8. Submission: Calendar, Demo, Video, Pitch and Final Output

8.1 Competition calendar and rules

Date	What happens
28 October	Full challenge statements and data lists are released. Re-read the statement and adjust.
2 November	Space Apps Connect opens for questions to NASA and mentors.
13 November	Global offers, the 2026 Project Submission Guide and the Judging and Awards Guide are released. Read both fully.
14 November, 9:00 a.m. local	The hackathon begins and project work starts.
15 November, 11:59 p.m. local	Submission closes. Submit well before this time.
16 November	Judging begins.
30 November	Participant survey is due.
Late December	Global winners are announced (in 2025 this was 18 December).

Local organisers may set earlier internal upload deadlines. The Bangladesh organiser's registration page showed 10 October 2026 as a closing date, so verify registration and the local timetable with the Local Lead now.

Rules to follow

- One challenge per team, at most six members, and every member registered and listed on the team page (confirm in the 2026 guide).
- Under-18 participants must be registered by a parent or guardian and accompanied by one at all times during the event (2026 Terms).
- No recognisable name, voice or likeness of a person under 18 may appear in a submission or video. Use screen recordings, illustrations and adult team members who agree to appear.
- **Open source:** original content must be freely available or under an OSI-approved licence (Apache 2.0 by default). Third-party fonts, icons, sounds and code must be freely available or open source, with licences listed. Cite every source, even open-source ones.
- The project must directly address the chosen challenge. No promotion of commercial products or services and no personal information in the submission.
- **Demo link:** for global judging the 2025 guide allowed one public link to a 30-second video or a 7-slide deck, and longer files were not eligible. Check the 2026 guide on 13 November.
- AI tools may be used, but state where they were used (confirm the wording in the 2026 guide).

Sources: https://www.spaceappschallenge.org/legal/ · https://www.spaceappschallenge.org/resources/team-formation-guide/ · https://www.spaceappschallenge.org/resources/project-submission-guide/

8.2 Three-minute judging demo and pitch

Time	What to say	What to show or do	What the judge should understand
0:00–0:20	The future of exploration depends on people understanding the systems that keep crews alive. Young learners need ways to practise these ideas, not just read about them.	Open on a power crisis, not the logo. One strong mission image and the exact 2026 challenge title.	Why the problem matters.
0:20–0:45	CrewReady turns the chosen NASA challenge and relevant open resources into a short cooperative outpost mission.	The exact official challenge and its mapping; a NASA source → game mechanic diagram.	Why the solution belongs in the hackathon.
0:45–1:30	Demonstrate one decision, the resource and safety consequence, and the source-based explanation.	Live: tap Play, handle one spacecraft obstacle, inspect systems, choose and see the state change.	The product works and decisions matter.
1:30–2:00	Show a twist or team clue and then the outcome. Show one brain-training activity and, if implemented safely, one optional body-coordination activity.	Twist or crew clue, then the outcome; the Training Deck screen, explained as learning mini-games, not validated tests.	Creativity, collaboration and replay value.
2:00–2:30	Open the science debrief and source drawer. Show the future discovery feed as a clearly labelled roadmap, not a completed feature.	Debrief and source drawer; a mock reel card with source attribution and captions.	Scientific validity and transparency.
2:30–3:00	Show a Quick Run retry and an upgrade preview. Explain who benefits, how a classroom or community could pilot it, what works today and what comes next. Close with impact, limitations and the next step.	Quick Run retry, upgrade preview, impact plan, public repository, source register and project page.	Retention and progression logic, and confidence without overclaiming.

The pitch statement

“CrewReady: Beyond Us is a junior astronaut mission trainer where young explorers command a lunar or Martian outpost and learn that survival is a systems problem. Power, water, air, food, radiation protection, crew time and trust compete for limited resources. A choice that saves one system can put another under pressure. Players investigate clues, manage uncertainty, support other crews at a real cost, and learn from every outcome through a clear science debrief. Built around authentic NASA resources and transparent simulation rules, CrewReady turns complex mission engineering into a game that is exciting without pretending space is easy.”

Judge questions to prepare for

- Which exact 2026 challenge are you answering, and how does the solution meet its requirements?
- What is NASA-sourced, what is a simplified model, and what is fictional game data?
- How did you check the rules and avoid misleading science?
- What happens if the internet drops or a player disconnects?
- Why would a child return after finishing the campaign?
- What did you deliberately leave out, and why?

8.3 Video concept and production scripts

Purpose: explain the problem, show the solution and make the experience memorable. This concept video does not replace a working demo. Confirm the organiser's actual duration, format and submission rules before final production.

- **Format:** 16:9 landscape with captions throughout. Make two cuts: a 30-second competition cut that fits the demo-link limit, and a 45–60 second concept cut for the local pitch and sharing.
- **Style:** cinematic Mars atmosphere mixed with clean animated UI; quick but readable transitions.
- **Narration:** calm, curious and slightly suspenseful. Use licensed music and original or credited visuals.
- If visuals are mockups, label them "Concept UI" or "Prototype visual." Do not imply unfinished features work.
- Do not use NASA branding in a way that suggests NASA endorsement. Credit NASA imagery and data.

The 30-second competition cut

Time	Visual / edit	Voice-over	On-screen text
0–4 s	The Mars craft in deep space; a radiation warning flashes and the dose meter climbs.	"A solar storm is coming, and your crew is far from home."	RADIATION WARNING
4–9 s	A thumb taps PLAY NOW; the spacecraft cutaway appears with four system rings.	"One tap starts your mission."	ONE TAP TO PLAY
9–17 s	The player compares three solution cards with costs, builds a shelter from stored supplies, and the rings update.	"Every fix costs something. Power, air, shielding and food are all linked."	EVERY CHOICE HAS A COST
17–24 s	The craft lands, the outpost lights return, and a science debrief shows a source label and a "game model" tag.	"Learn the real science behind every outcome, then launch again."	REAL SCIENCE. CLEARLY LABELLED SIMULATION.
24–30 s	Four role cards, the title and the tagline.	"Solo or with your crew. CrewReady: Beyond Us."	CREWREADY: BEYOND US

The concept cut (about 58 seconds)

Time	Visual / edit	Voice-over	On-screen text
0–5s	Mars outpost at dusk; lights flicker; battery indicator falls.	"Imagine you're responsible for keeping a space outpost alive. Then the sunlight begins to disappear."	POWER DROPPING
5–12s	Dust sweeps over solar panels; power, food and life-support indicators react.	"One storm can create a chain of decisions. Save power now, or protect the systems your crew will need later?"	ONE EVENT. CONNECTED SYSTEMS.
12–20s	CrewReady home screen; thumb taps PLAY NOW; Nova introduces Mission 01.	"Meet CrewReady: Beyond Us, a space survival game where every choice changes the mission."	ONE TAP TO PLAY
20–31s	Player chooses battery upgrade, checks sensor warning, reroutes power; dashboard changes.	"Build your outpost, manage limited supplies, investigate warnings, and adapt when the unexpected hits."	BUILD · DECIDE · SURVIVE

Time	Visual / edit	Voice-over	On-screen text
31–41s	Four role cards on separate phones; players share preset clues; crisis resolves.	“Go solo or bring your crew together. Different clues, shared decisions, one mission to keep running.”	SOLO OR CREW UP
41–51s	Outcome and science debrief show a source and “game model” label.	“Then discover the science behind the outcome, learn from your choices, and launch again.”	REAL SCIENCE. CLEARLY LABELLED SIMULATION.
51–58s	Outpost lights return; title and tagline.	“CrewReady: Beyond Us. One tap to play. A universe of missions to master.”	CREWREADY: BEYOND US

Production checklist

- Confirm deadline and late-submission policy before polishing.
- Prepare seven scenes as clips or animated stills, one idea per scene.
- Record a clear voice-over or properly licensed synthetic voice.
- Use captions and high-contrast text; do not rely on audio alone.
- Use original/licensed assets and properly credited imagery.
- Export 1080p MP4 unless official instructions specify otherwise; test on a phone.
- Avoid private contact details and keep the clean export plus project file.
- **Check every frame and the audio:** no recognisable name, voice or face of anyone under 18, and adults who appear must agree.
- Confirm the demo link is public with no login and that the competition cut ends at 30 seconds.

8.4 Final output checklist

Deliverable	Owner	Done when
Working playable demo	Developer	The public link works with no login, the main path completes and a backup recording exists.
Public repository	Developer	A readable README with setup and run instructions, licence, source register and credits; tested in a private window.
NASA project page	Product lead	Every required field is complete, with the challenge title and its core ask, data sources, links, team and media, before the deadline.
30-second video and concept video	Story/pitch lead	Captions, licensed assets, no under-18 likeness, and tested on a phone.
Three-minute pitch and backup	Story/pitch lead	Rehearsed, with the judge questions prepared.
Source and assumption register	Science/data lead	Every value has a source or a game-model label and the in-app source drawer works.
User-test notes	QA/integration	3–5 unfamiliar testers, one trade-off explained by each, hesitation points fixed, nobody identifiable.
Submission confirmation	Product lead	Submitted before the deadline and re-checked from a logged-out browser.

9. Risks and Safeguards

Risk / problem	Fix
Ten-day mission loses players	Campaign stages are short sessions, not real-time wait timers. Quick Run exists from day one.
Quick Run feels repetitive	Curated event combinations, modifiers, upgrades and daily seeded challenges; keep outcomes explainable.
Random events feel unfair	Clues, bounded difficulty, viable strategies and a post-event explanation.
Multiplayer overwhelms team	Solo first; test networking early; room codes and preset messages; solo fallback.
Same-Wi-Fi join unreliable	Do not promise automatic LAN discovery until validated on target phones.
Reels create moderation risk	Defer uploads until moderation, reporting, rights and safeguarding are ready.
Too many unrelated features	Every feature must strengthen mission play, science learning or retention.
Science looks invented	Source register, visible labels, separate game estimates.
Live API breaks demo	Cached timestamped data or clearly labelled local scenario.
Game too hard for younger players	Short tutorial, optional hints, simple mode if time allows, plain-language debrief.
Rewards become manipulative	No streak shame, loot boxes, energy timers that block play or pay-to-win.
NASA endorsement implied	Use factual attribution; never imply NASA sponsors or endorses CrewReady.
Spacecraft obstacles contain unsourced science	Every obstacle card links to a NASA source and stays marked verify until checked; values are labelled as game units.
Video is longer than the allowed demo length	Make a 30-second competition cut and keep the longer cut for the local pitch.
Work starts before the official start time	Keep preparation to registration, reading, planning and tool practice; build from 14 November.

10. Future Roadmap: Content, Creators and Platform

Landscape space reels can become a later discovery layer, but they should not become the main product. The distinctive loop is play → decide → see consequences → understand the science → replay or upgrade.

Space Reels (working title) is a future growth layer, not a hackathon dependency: a vertical-screen, horizontal-swipe feed where each card is a short, useful discovery. The interaction can feel familiar to young users, but the editorial standard should be closer to a mini science museum than an endless attention trap. It may contain short science explainers, NASA-data visualisations, mission breakdowns, experiment demonstrations, creator challenges and beautiful space visuals. A viewer can swipe to the next clip, save it, follow a topic, or launch a related playable mission. Each clip should connect back to a source or learning objective where applicable.

Feature	Potential value	Requirements before launch
Space Reels	Short landscape explainers about space, engineering and discoveries.	Curated/moderated content, captions, source credits and age-appropriate review.
Creator submissions	Educators, illustrators and science communicators submit explainers.	Review workflow, copyright verification, reporting and moderation capacity.
Creator rewards	Potential incentives for approved contributors.	Legal review, transparent rules, funding and safeguarding; no promises until operational.
Discover Library	Save and revisit science cards.	Source quality, accessibility and regular review.
Teacher/family mode	Replay missions and discuss trade-offs.	Privacy-by-design, no unnecessary student accounts and educator testing.
Earth Detective	Explore credited satellite imagery and compare environmental change.	Real dataset and a clear match to an official challenge.

10.1 Space Reels: reel types and feed rules

Reel type	Example	Learning / product purpose
NASA data visual	A solar storm timeline, lunar surface temperature map, or Earth observation comparison.	Turns real data into a visual question and links back to the source.
One-minute science	Why a habitat needs water recovery, how solar panels generate power, or why dust matters.	Explains one idea with captions, diagrams, and a source note.
Mission replay	A creator or player shows a choice, the result, and what they would change.	Builds reflection and peer learning.
Interactive visual	Swipe to compare safe and unsafe power levels or rotate a habitat module.	Invites active exploration rather than passive watching.
Creator challenge	“Explain this NASA chart in 3 frames” or “Design a better lunar shelter.”	Encourages original science communication and community contribution.

Feed rules for retention without dark patterns

- Every reel must deliver one clear idea, a question, or a useful visual. Avoid filler, fake urgency, misleading claims, or infinite autoplay by default.

- Add captions, readable source labels, alt text where practical, and a “Learn more” route to the NASA source or project explanation.
- Use a finite session, clear exit, optional reminders, and a “continue mission” action. No streak-loss guilt or pressure to keep scrolling.
- Use a simple recommendation system based on chosen interests and completed topics only if consent and privacy are handled appropriately. Offer chronological or topic-based browsing as an alternative.
- **MVP boundary:** no reels feed in the core judging build unless the mission simulator is already complete. A static mock-up, a small curated swipeable set or a roadmap slide is enough. Do not build a full recommendation algorithm, creator economy or moderation system during the hackathon.
- **Retention loop:** finish a mission → receive a science debrief → unlock a relevant short explainer → try a challenge → return to the next mission. The reel is an optional bridge to learning, not a replacement for gameplay.
- **Quality controls:** creator attribution, source links, age-appropriate moderation, content reporting, accessibility captions, clear distinction between fact and speculation, and a review process for scientific claims.

10.2 Creator ecosystem and earning potential

The long-term opportunity is a creator layer where educators, students, illustrators, animators and science communicators publish short explainers or interactive visuals. In a mature product, approved contributors could potentially be rewarded through educational partnerships, commissioned content, grants, sponsorships, classroom partnerships, school licences or a transparent revenue-sharing programme. This is a product hypothesis, not a guaranteed income promise, and it is not part of the hackathon MVP. Do not promise that users will earn money in the current product, and do not introduce a public creator economy, ads, direct messaging or open comments during the hackathon.

Before enabling payments or open uploads, the team would need age-appropriate safeguards, rights and attribution checks, NASA data and source verification, moderation, privacy protections, child-safety review and a clear policy for AI-generated material. For minors, participation and monetisation would require additional guardian and legal review, appropriate adult involvement, consent, transparent eligibility, fraud controls and compliance with relevant law and platform policies. Do not launch open comments, direct messaging or public creator payments in the initial version.

11. Research and Source Register

- **Official challenge:** Build a Junior Astronaut Mission Trainer, NASA Space Apps 2026. https://www.spaceappschallenge.org/2026/challenges/build-a-junior-astronaut-mission-trainer/
- NASA Space Apps 2026 theme and event information; re-check current requirements. https://www.spaceappschallenge.org/2026/ and https://science.nasa.gov/universe/nasa-welcomes-all-ages-experience-levels-for-space-apps-challenge/
- NASA Space Apps 2026 Terms and Conditions and Team Formation Guide. https://www.spaceappschallenge.org/legal/ and https://www.spaceappschallenge.org/resources/team-formation-guide/
- **NASA 2025 Global Winners:** PureFlow, HerCode Space, Zumorroda-X, Resonant Exoplanets, Twisters. https://www.nasa.gov/learning-resources/stem-engagement-at-nasa/nasa-announces-2025-international-space-apps-challenge-global-winners/
- **NASA 2024 Global Winners:** Eco-Metropolis (Innovisionaries). https://www.nasa.gov/learning-resources/stem-engagement-at-nasa/nasa-international-space-apps-challenge-announces-2024-global-winners/
- **NASA 2023 Global Winners:** Team Voyagers and storytelling and data examples. https://science.nasa.gov/directorates/smd/2023-nasa-international-space-apps-challenge-announces-10-global-winners/
- Bangladeshi global winners, as reported by The Business Standard. https://www.tbsnews.net/bangladesh/27-bangladesh-teams-vie-nasa-space-apps-challenges-2024-959676
- NASA Environmental Control and Life Support Systems (ECLSS): atmosphere, oxygen, ventilation, waste and water. https://www.nasa.gov/reference/environmental-control-and-life-support-systems-eclss/
- **NASA Human Research Program, Space Radiation:** background for radiation hazard explanations. https://www.nasa.gov/hrp/hazard-space-radiation/
- NASA Orion radiation protection plan (shelter built from stowage bags). https://www.nasa.gov/missions/artemis/orion/scientists-and-engineers-evaluate-orion-radiation-protection-plan/
- **NASA Space Place, What Powers a Spacecraft?:** child-friendly spacecraft power and solar-panel dust explanation. https://spaceplace.nasa.gov/what-powers-a-spacecraft/en/
- **Earth–Mars communication delay:** NASA ASCEND paper on delays, disruptions and blackouts for crewed Mars missions. https://ntrs.nasa.gov/citations/20220013418
- **Common Sense Media, The 2025 Common Sense Census:** Media Use by Kids Zero to Eight. https://www.commonsensemedia.org/research/the-2025-common-sense-census-media-use-by-kids-zero-to-eight
- Pew Research Center, Teens, Social Media and Technology 2024. https://www.pewresearch.org/wp-content/uploads/sites/20/2024/12/PI_2024.12.12_Teens-Social-Media-Tech_REPORT.pdf
- Pew Research Center, Teens and Video Games Today, 2024. https://www.pewresearch.org/internet/2024/05/09/teens-and-video-games-today/

12. Conclusion: Why CrewReady Can Win

Strength	Why it scores	Evidence we will show
Exact challenge fit	Relevance is the first filter.	Every mechanic maps to outpost life support, radiation shielding, power and food, and the challenge title appears in every submission item.
Science that changes play	Validity and NASA data usage.	Source-linked obstacles: radiation shelters, communication delay, CO_2 scrubbing and dust all change the player's options, and a source drawer is one tap away.
One tap start, endless return	User experience, Impact and retention.	Quick Run starts in about 25 seconds, cargo improves the outpost, and a Daily Voyage gives a reason to come back.
Lessons from previous winners	Creativity and storytelling.	Threat testing like PureFlow, narrative like HerCode Space, and mini-games only where they teach a mission mechanic.
Teamwork and trust	Creativity and Teamwork.	Crew roles with different clues, costly resource transfers and a solo fallback.
Honest scope	Presentation and credibility.	A narrow, tested MVP, a clearly labelled roadmap, no overclaiming, and a backup recording of the demo.

Build the smallest complete product that visibly solves the chosen 2026 challenge. The training activities should deepen the learning loop. The horizontal reel feed and creator earnings model should remain future roadmap items unless the core mission is already stable. Do not claim a feature is implemented unless a judge can use it in the submitted prototype.

CrewReady's strongest promise is not that it will turn every child into an astronaut. It is that it can help young people practise curiosity, systems thinking, teamwork, and evidence-based decisions while exploring the science behind humanity's next frontier.

> **FINAL PRODUCT PRINCIPLE**
> MAKE SURVIVAL REAL. MAKE THE SCIENCE VISIBLE. MAKE EVERY CHOICE MATTER.

BUILD LESS. PROVE MORE. MAKE THE SCIENCE MATTER.